Compensation Type SOP: Borrower Paid vs. Lender Paid

Feature Overview: Compensation Type Logic

The pricing engine must handle two distinct compensation structures. **Lender Paid** compensation is built into the interest rate, while **Borrower Paid** compensation is stripped from the rate and charged as a separate fee.

- **Why it's there**: Borrower Paid structures are used to achieve lower interest rates or to satisfy specific anti-steering and high-cost testing requirements.
- **Pricing Engine Impact**: Switching to Borrower Paid requires the engine to "Thin" the raw lender pricing by subtracting the company's fixed LO Comp Plan and re-calculating the final cost to the borrower in Section A.

Following the Mortgage Pricing Engine Logic Guide, the following scenario-based thresholds apply:

Comp Type	Pricing Action (Math)	Fee Sheet Impact (Section A)
Lender Paid	Raw Price + 0.000	$0.00 (Built into Rate)
Borrower Paid	**Raw Price - Default Comp (1.25%)**	**0.75% of Loan Amount**

Developer Logic: If comp_type == "Borrower Paid," the system must execute the following logic sequence:

1. **Strip Company Comp**: Identify the LO's fixed plan (1.25%). This value must be **subtracted** from the raw price returned by Mortech Marksman.

2. **Apply Borrower Fee**: Enable the custom fee field. The user has selected 0.75% (75 bps). This must be calculated as a dollar amount and mapped to **Section A**.

How to Pull This Information from the App

The engine monitors the Compensation Type dropdown and the specific percentage inputs.

Requirement	Description
Data Source	COMPENSATION node in the LoanApp UI.
Prefill Field 1	lo_comp_plan_default (Fixed at **1.250** for this scenario).
Prefill Field 2	borrower_paid_fee_percent (Entered as **0.750**).
Prefill Field 3	loan_amount (The base for the dollar calculation).

Mathematical Walkthrough (Scenario)

1. Identify Variables

- **Mortech Raw Price**: 103.000 (3.000 Gain)
- **Lender Paid Comp (Default)**: 1.250%
- **Selected Type**: Borrower Paid
- **Custom Borrower Fee**: 0.750% (75 bps)
- **Loan Amount**: $400,000

Step A: Price Adjustment (The "Thinning" Process)

- Formula: Raw Price - Default Comp
- $$103.000 - 1.250 = 101.75$$
- **Result**: The borrower sees a price of 101.750 (a credit of 1.750 points).

Step B: Section A Fee Calculation

- Formula: Loan Amount x (Borrower Comp Fee / 100)
- $$400,000 \times 0.0075 = \$3,000.00$$
- **Result**: $3,000.00 is mapped to **Section A (Origination Charges)**.

Logic Filters

- **Pricing Modification**: If comp_type == "Borrower Paid", then final_display_price = raw_price - 1.25.

- **Fee Generation**: If comp_type == "Borrower Paid", calculate origination_charge = loan_amount * 0.0075 and place in Section A.

Implementation Code

Node.js Implementation

```
/**

 * Calculates the final price and triggers APR recalculation for
Borrower Paid compensation.

 * * @param {Object} priceData - Includes rawPrice, loanAmount, and
interestRate.

 * @param {Object} compConfig - The compensation configuration (e.g.,
type: 'Borrower Paid').

 * @param {Array} existingFees - The current array of closing costs on
the loan.

 */

function calculateCompScenario(priceData, compConfig, existingFees) {

  const { rawPrice, loanAmount, interestRate } = priceData;

  const LENDER_PAID_DEFAULT = 1.250; // Company standard margin

  const BORROWER_PAID_FEE = 0.750;  // User selection (75 bps)



  let finalPrice = rawPrice;

  let sectionAFee = 0;

  let updatedFees = [...existingFees]; // Clone the existing fee array
to avoid mutating state directly

  let newAPR = null;



  if (compConfig.type === 'Borrower Paid') {

    // 1. "Thin" the Price: Subtract the 1.25% Lender Paid Comp

    finalPrice = rawPrice - LENDER_PAID_DEFAULT;
```

```
    // 2. Calculate the 0.75% fee for Section A
    sectionAFee = loanAmount * (BORROWER_PAID_FEE / 100);


    // 3. Inject the Section A fee as a Prepaid Finance Charge (PFC)
    updatedFees.push({
      name: "Borrower Paid Compensation",
      amount: sectionAFee,
      section: "Section A",
      isPFC: true // CRITICAL: This flag tells the APR engine to
deduct this from the Amount Financed
    });


    // 4. Trigger APR Recalculation
    // (Passing the newly updated fee array into your system's APR
utility)
    newAPR = calculateAPRUtility({
        loanAmount: loanAmount,
        interestRate: interestRate,
        fees: updatedFees
    });
  }


  // 5. Separate outputs for UI vs. Mortech XML
  return {
    uiPayload: {
        displayPrice: finalPrice,              // Shown to user as
```

```
points/credits
        sectionAFeeAmount: sectionAFee,        // Mapped to Estimated
Closing Costs
        updatedAPR: newAPR                     // Refreshed APR value
    },
    mortechXMLPayload: {
        priceMark: finalPrice                  // MUST be the thinned
price, not raw
    }
  };
}


/**
 * Mock APR Utility to illustrate how the fee impacts the math
downstream.
 */
function calculateAPRUtility(loanDetails) {
    // Isolate only the fees that affect the APR (Prepaid Finance
Charges)
    const totalPFCs = loanDetails.fees
        .filter(fee => fee.isPFC)
        .reduce((sum, fee) => sum + fee.amount, 0);


    // Calculate the true Amount Financed
    const amountFinanced = loanDetails.loanAmount - totalPFCs;


    // [Complex APR amortization math goes here using amountFinanced]
```

```
    return "7.125"; // Mock return value
}
```

Technical Integration Notes

- **Precision**: All percentage math must be handled to 3 decimal places (e.g., 0.750) to prevent rounding errors in Mortech XML transmissions.
- **UI Toggle**: If "Borrower Paid" is toggled, the UI must immediately refresh the price list and show the $3,000 fee in the estimated closing costs.
- **Mortech XML**: Ensure the Price node in the outgoing request reflects the "Thin" price after the 1.25% subtraction.

Developer Supplement: Recalculating Loan Price and APR

When a user toggles the compensation type from **Lender Paid** to **Borrower Paid**, the pricing engine must simultaneously adjust two distinct financial metrics: the **Loan Price** (points/credits) and the **APR** (the total cost of the loan expressed as a yearly rate).

Here is the data flow and logic required for both.

1. Recalculating the Loan Price (The "Thin" Price)

The Loan Price calculation is straightforward but critical for the Mortech integration. When shifting to Borrower Paid, the company's default compensation must be stripped out of the raw pricing. Because everyone's compensation is different, the user should enter this at the time of selecting BP pricing. See the ARIVE image below:

- **Trigger:** `comp_type == 'Borrower Paid'`
- **Logic:** Subtract the static company margin (`lo_comp_plan_default`) from the incoming Mortech mark.
- **Formula:** `Final Display Price = Raw Price - 1.250 (1.25 is ex only)`
- **Developer Action Item:** The resulting `Final Display Price` is what must populate on the front-end UI for the borrower, and crucially, this is the exact value that must be injected into the `Price` node of the outgoing Mortech XML payload. **Do not send the raw price.**

2. Recalculating the APR (Impact of Section A Fees)

While the SOP covers calculating the Section A Origination Charge, the dev team must ensure this new fee correctly routes into the **APR calculation engine**.

APR is dependent on the *Amount Financed* (Loan Amount minus Prepaid Finance Charges). Because the new Borrower Paid compensation fee is mapped to Section A, it acts as a **Prepaid Finance Charge (PFC)**.

- **Trigger:** Generation of the new `sectionAFee`.
- **Logic:** The calculated dollar amount (e.g., $3,000) must be flagged as `APR Affecting` (or `PFC = true`) within your fee payload.
- **Data Flow:**
    1. System calculates `sectionAFee = loanAmount * (borrower_paid_fee_percent / 100)`.
    2. System adds `sectionAFee` to the array of existing Prepaid Finance Charges (e.g., underwriting fees, processing fees, points).
    3. System calculates the new Amount Financed: `Amount Financed = Loan Amount - (Total Existing PFCs + sectionAFee)`
    4. System triggers the existing APR calculation utility using the *new* Amount Financed and the selected interest rate.
- **Developer Action Item:** Ensure the state management seamlessly passes the newly generated `sectionAFee` into the APR calculation function before the UI refreshes. If the fee is added to the UI but omitted from the APR payload, the system will face compliance failures.

Summary of System Events (Execution Order)

To prevent race conditions or UI lag, developers should execute the state updates in this specific order when the user selects "Borrower Paid":

1. **Calculate Final Price:** `rawPrice - 1.250`
2. **Calculate Section A Fee:** `loanAmount * 0.0075`
3. **Update Fee Array:** Inject the Section A fee into the Closing Costs array, ensuring it is flagged as an APR-affecting Prepaid Finance Charge.
4. **Trigger APR Recalculation:** Run the APR function using the updated fee array.
5. **Render UI & XML:** Push the updated Price, Closing Costs, and APR to the frontend, and format the Mortech XML.

BELOW ARE EXOF HOW FEE SHEETS SHOULD LOOK, AND YOU CAN UNDERSTAND SECTION A - LENDER COSTS, AND SECTION B - THIRD PARTY SERVICES

Closing Cost Details

Loan Costs

A. Origination Charges	**$1,802**
.25 % of Loan Amount (Points)	$405
Application Fee	$300
Underwriting Fee	$1,097

B. Services You Cannot Shop For	**$672**
Appraisal Fee	$405
Credit Report Fee	$30
Flood Determination Fee	$20
Flood Monitoring Fee	$32
Tax Monitoring Fee	$75
Tax Status Research Fee	$110

C. Services You Can Shop For	**$3,198**
Pest Inspection Fee	$135
Survey Fee	$65
Title – Insurance Binder	$700
Title – Lender's Title Policy	$535
Title – Settlement Agent Fee	$502
Title – Title Search	$1,261

D. TOTAL LOAN COSTS (A + B + C)	**$5,672**

Other Costs

E. Taxes and Other Government Fees	**$85**
Recording Fees and Other Taxes	$85
Transfer Taxes	

F. Prepaids	**$867**
Homeowner's Insurance Premium (6 months)	$605
Mortgage Insurance Premium (months)	
Prepaid Interest ($17.44 per day for 15 days @ 3.875%)	$262
Property Taxes (months)	

G. Initial Escrow Payment at Closing	**$413**
Homeowner's Insurance $100.83 per month for 2 mo.	$202
Mortgage Insurance per month for mo.	
Property Taxes $105.30 per month for 2 mo.	$211

H. Other	**$1,017**
Title – Owner's Title Policy (optional)	$1,017

I. TOTAL OTHER COSTS (E + F + G + H)	**$2,382**

J. TOTAL CLOSING COSTS	**$8,054**
D + I	$8,054
Lender Credits	

Calculating Cash to Close

Total Closing Costs (J)	$8,054
Closing Costs Financed (Paid from your Loan Amount)	$0
Down Payment/Funds from Borrower	$18,000
Deposit	– $10,000
Funds for Borrower	$0
Seller Credits	$0
Adjustments and Other Credits	$0
Estimated Cash to Close	$16,054

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Quote Number: 10709057

Preparation Date: 03/29/2026 07:16 PM

Loan Purpose:	Purchase	Purchase Price:	$439,000.00	Total Loan Amount:	$431,048.61
Property Type:	Single Family (1-4 Units)	Occupancy:	Primary Residence	No. of Units:	1
Credit Score:	Verified	ZIP / State:	95926 / California	Escrow:	None Waived
Product:	First Colony Wholesale - FHA 30 Year...	Rate / APR:	5.625% / 6.437%	Lock Period:	30 Days

Lender Fees — $1,780.36

0.130% of Loan Amount (Points) ⚠	$560.36
Underwriting Fee	$1,220.00

Third Party Fees — $13,414.61

Services You Cannot Shop For	
Appraisal Fee	$795.00
Credit Report Fee	$105.00
FHA Upfront MI (Financed) ⚠	$7,413.61
Flood Certification Fee	$8.00
Lender Wire Fee	$75.00
Processing Fee (3rd Party)	$1,395.00
Tax Monitoring Fee	$75.00
Tax Status Research Fee	$110.00
Services You Can Shop For	
Title - Insurance Binder	$700.00
Title - Settlement Agent Fee	$502.00
Title - Title Search	$1,261.00
Title Escrow/Settlement Fee	$975.00

Taxes and Other Government Fees — $200.00

⊞ Recording Fees	$200.00
⊞ Transfer Taxes	$0.00

Prepaids and Initial Escrow Payment at Closing — $4,506.96

Prepaids	
Hazard Insurance Premium(12 Months @ $115.24)	$1,382.85
Mortgage Insurance Premium(__ Months @ $193.01)	--
Prepaid Interest(3 Days @ $66.4287)	$199.29
Property Taxes(3 Months @ $429.85)	$1,289.55
Supplemental Property Insurance Premium(__ Months @ __)	--
Initial Escrow Payment at Closing	
Aggregate Adjustment	--
Hazard Insurance Reserve(3 Months @ $115.24)	$345.72
Mortgage Insurance Reserve(__ Months @ $193.01)	--
Property Taxes(3 Months @ $429.85)	$1,289.55
Supplemental Property Insurance Reserve(__ Months @ __)	--

Estimated Proposed Monthly Housing Expense

First Mortgage P&I	$2,481.36
Other Financing P&I	$0.00
Homeowner's Insurance	$115.24
Supplemental Property Insurance	$0.00
Property Taxes	$429.85
Mortgage Insurance	$193.01
Homeowner Assn. Dues	$0.00
Other	$93.00
TOTAL ESTIMATED MONTHLY PAYMENT	$3,312.45

Estimated Funds To Close

Downpayment/Funds from Borrower	$15,365.00
Lender Fees	$1,780.36
Third Party Fees	$13,414.61
Taxes and Other Government Fees	$200.00
Prepaids and Initial Escrow	$4,506.96
Estimated Total Payoffs	$0.00
Funds Due From Borrower (A)	**$35,266.93**
Deposit	$0.00
Lender Credits	$0.00
Seller Credits	$0.00
Adjustments and Other Credits	$0.00
FHA UFMIP / VA Fee / MI Financed	$7,413.61
Subordinate Financing	$0.00
Total Credits Applied (B)	**$7,413.61**
CASH FROM BORROWER (A-B)	$27,853.32